

and are used to communicate with CSG members. Even if the enemy intercepts military communication, they will not be able to understand the message.

VHF is used for ship-to-ship or helicopter-to-ship communications. Military HF and MF have a range of around 300km. VHF and UHF, due to their high-frequency-induced attenuation, have a range of around 100km.

Non-military radio communication is not secure. It is used to communicate with harbours, ships and aircraft of civilian or neutral origin. Non-military UHF systems can be used up to a range of 50km; all other band systems have the same range as that of their military counterparts.

**Satellite communication systems.** Aircraft carrier, being the commanding ship of the CSG, has a variety of dedicated satellite communication systems for various purposes. Long-distance communications are carried out through the USA's defence satellite communications system (DSCS), a dedicated satellite based communication network. This high-capacity global system uses SHF band geostationary communication satellites that orbit equidistantly around the world.

On the other hand, short-distance communications are managed by fleet satellite communication (FLTSATCOM). Typically, this type of communication is between two CSGs or intra CSG communication or shore-to-CSG communication.

Satellite communication is carried out in two modes: receipt and broadcast. Receipt mode requires the recipient to send an acknowledgement message for each message it receives. Whereas, broadcast method, also called do-not-answer mode, does not require an acknowledgement message.

In the latter mode, the CSG can

keep on receiving the data that are sent to it. The great advantage is that the CSG can preserve radio silence. Because, the moment it sends an acknowledgement message, the enemy electronic support measure (ESM) systems can find the CSG's location. Through this mode, the CSG can receive situational updates without revealing its position.

### **Tactical digital information links (TADIL)**

These tactical datalinks are the bridge between C<sup>2</sup> systems of individual ships. Through these datalinks, Aegis ships exchange data between them and also with the carrier, so that overall C<sup>2</sup> is maintained.

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HMS Sheffield—a destroyer of the UK—was sunk during Falklands War by Argentine aircraft launching French-made Exocet anti-ship missiles. It is said that at the time of attack, the ESM module of the EW system was not on due to interference problems with satellite radios. Had the ESM been on, perhaps, that ship could have survived the attack

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**Link-4A.** When airborne early warning (AEW) aircraft detect incoming enemy bomber, F-18 aircraft on patrol are instructed to fly towards the general area of the incoming enemy bomber. On reaching the area, these are vectored (giving directional commands) to intercept the bomber.

These are vectored through a datalink. The aircraft computer displays in the pilot's head-up display

(HUD) shows the waypoints to be followed to intercept the bomber.

Link-4A is one such data link. It is a non-secured two-way datalink used for providing vector commands to combat aircraft. It operates in UHF band at a data rate of 5kbps. It follows time-division techniques. Also, aircraft are communicated during the following processes: automatic carrier landing system (ACLS), air traffic control (ATC) and during combat strikes.

Generally, Link-4A's transmissions are not secure and could be jammed by enemy ECM. On the positive side, it is considered easier to operate and maintain. Because of these reasons, this very old datalink is still being used by the US Navy. In a carrier, there are ten channels for this link and the typical range is as far as 500km.

**Link-11.** Link-11 is a half-duplex high-speed encrypted datalink. It operates in HF and UHF bands. It facilitates computer-to-computer digital data communication between aircraft and ships. Even if the signals are intercepted by the enemy, these cannot be understood by them because of the encryption. Data can only be decoded at the receiving terminal.

Positional information of the transmitter is also sent along the data so that the receiver can know the sender's position. This feature is an important aspect of combat communication. Typically, data sent by a ship to the aircraft is received by appropriate receivers in the aircraft. These are decoded by decoders and sent to the aircraft computer.

Then, information is displayed to the pilot in a format that he or she can understand. Data can be anything like waypoints that have to be followed for reaching the new destination or positional data of the enemy combat aircraft.

Combat aircraft have an